Prenatal Substance Exposures and Childhood Cognitive Performance:

A Multivariable Analysis

Abstract

This study examined associations between prenatal and perinatal substance exposures and childhood cognitive performance in 1,293 mother–child dyads. Using linear regression, we assessed tobacco, alcohol, opioid, and cannabis exposures in relation to cognitive score. In multivariable analyses, maternal smoking before and during pregnancy, prenatal opioid exposure, and prenatal cannabis exposure were independently associated with lower cognitive scores, while alcohol exposure and biospecimen-based markers were not. Stepwise selection using AIC and BIC retained the same set of predictors, suggesting stability of the selected model within this dataset and candidate set. These findings suggest that specific prenatal substance exposures are associated with poorer childhood cognitive performance.

1 Introduction

Prenatal exposure to substances such as tobacco, alcohol, opioids, and cannabis is a public health concern because these substances can cross the placenta and affect fetal neurodevelopment. Prior studies have linked prenatal substance exposure to adverse cognitive outcomes in offspring, but most research focuses on individual exposures and relies primarily on self-reported measures. Fewer studies integrate biomarker-based assessments or examine the combined effects of multiple exposures.

This study examines the associations between prenatal and perinatal substance exposures—including smoking before and during pregnancy, alcohol, opioid, and cannabis use—and childhood cognitive performance. Both questionnaire-based measures and biospecimen markers are considered, along with a poly-substance exposure index. We assess these associations using regression models designed for both explanatory and predictive purposes.

2 Methodology

2.1 Data Exploration

Data exploration did not identify implausible values. Missingness varied across variables, with questionnaire-based exposures exhibiting minimal missing data and biospecimen-based measures showing low levels of missingness (Fig. 1), consistent with sample non-collection or assay failure rather than data quality issues.

The outcome variable (*cog_score*) was approximately normally distributed and analyzed on its original scale (Fig. 2). In contrast, cord blood cotinine exhibited pronounced right-skewness (Fig. 2) and was ***log(x + 0.1)***–transformed to stabilize variance and reduce the influence of extreme values.

Several exposure variables were aligned by design (e.g., *prenatal_alcohol* and *alcohol_preg*), resulting in substantial collinearity; therefore, aligned variables were not entered simultaneously into the same models. Because the poly-substance exposure index is a deterministic function of individual substance exposures, it was analyzed separately from its component variables to avoid perfect collinearity.

For biospecimen-based variables with missing data, cognitive score distributions were compared between individuals with and without available measurements, and no systematic differences were observed (Fig. 3). Given adequate remaining sample size, missing data were handled using listwise deletion.

2.2 Data Analyses

For both univariate and multivariate analyses, binary exposure variables were modeled as categorical predictors, while continuous variables were modeled on their numeric scale. Since the outcome variable (*cog_score*) is continuous and approximately normally distributed, associations were modeled using linear regression rather than generalized linear models.

Univariate linear regression models were fitted for each exposure. For binary exposures, the intercept represents the mean cognitive score in the unexposed group, while the slope represents the mean difference associated with exposure.

Multivariable linear regression models were used to estimate the independent associations between multiple exposures and cognitive score. All exposure variables were initially considered, including questionnaire-based measures, biospecimen-based markers, and continuous biomarkers. Since the poly-substance exposure index is a deterministic function of individual substance exposure indicators, it was not included simultaneously with its component exposure variables to avoid collinearity and non-identifiability of regression coefficients.

In addition to explanatory models, a predictive model was evaluated to assess out-of-sample performance. The final model structure was refitted on a training subset (70%) and evaluated on a held-out test set (30%) using root mean squared error (RMSE). As a sensitivity analysis, a Bayesian predictive model was also fitted using weakly informative priors and Gibbs sampling; out-of-sample performance was essentially unchanged, so the primary predictive is used and Bayesian details are provided in Appendix A.

2.3 Model Selection

Model selection was primarily guided by stepwise procedures based on AIC and BIC, while also considering scientific plausibility, global and partial F-tests, model diagnostics, and coefficient stability. It was informed by scientific plausibility, interpretability, and identifiability considerations.

2.4 Model Checking and Diagnostics

In Fig. 4, model assumptions were evaluated using standard diagnostic procedures for linear regression. Residual-versus-fitted plots were examined to assess linearity and homoscedasticity. Normal Q–Q plots were used to evaluate the approximate normality of residuals, with primary emphasis placed on visual assessment rather than formal hypothesis testing due to the large sample size.

Potential outliers and influential observations were assessed using leverage and Cook's distance diagnostics; no single observation was found to exert undue influence on model estimates. Multicollinearity was evaluated using variance inflation factors (VIFs), and all retained predictors were below commonly used thresholds, indicating no problematic collinearity in the final selected models.

3 Results

3.1 Study Sample and Exposure Characteristics

A total of 1,293 mother–child dyads were included in the final analytic sample. The cognitive score (cog_score) was approximately normally distributed, with a mean (SD) of 100.2 (14.7).

Prenatal substance exposures were common. Maternal smoking prior to pregnancy and during pregnancy were reported by 30.4% and 25.8% of participants, respectively. Alcohol use during pregnancy occurred in 14.8% of the sample. Prenatal opioid and cannabis exposures were observed in 7.8% and 15.5%, respectively. Biospecimen-based urine markers indicated opioid exposure in 11.4% and cannabis exposure in 18.4% of participants.

3.2 Univariate Analyses

In univariate analyses, several prenatal exposures were linked to lower cognitive scores. Maternal smoking, both before and during pregnancy, showed a significant negative association, with stronger effects during pregnancy. Prenatal opioid and cannabis exposures were also associated with lower scores, whereas alcohol exposure showed no significant effect. Overall, each additional prenatal substance exposure corresponded to an average 3.5-point decrease in cognitive score, and higher cord blood cotinine levels were inversely associated with cognition in unadjusted analyses.

(Results summarized in Table 1 and illustrated in Figures 1–3.)

3.3 Multivariable Analyses

In multivariable models adjusting for all candidate exposures, maternal smoking prior to pregnancy and smoking during pregnancy remained independently associated with lower cognitive scores. Smoking prior to pregnancy was associated with an approximately 2.9-point reduction, and smoking during pregnancy with an approximately 4.7-point reduction, in cognitive score.

Prenatal opioid and prenatal cannabis exposures also remained significant, corresponding to reductions of approximately 5.4 points and 4.0 points, respectively. Prenatal alcohol exposure, urine toxicology markers, and cord blood cotinine levels were no longer independently associated with cognitive score after adjustment.

The full multivariable model was statistically significant overall ($p < 0.001$).

(Results summarized in Table 2.)

3.4 Model Selection and Robustness of Findings

Backward stepwise selection using AIC and BIC, and the Bayesian model yielded identical final models, retaining smoking prior to pregnancy, smoking during pregnancy, prenatal opioid exposure, and prenatal cannabis exposure. Estimates were consistent with those from the full model, indicating robust findings.

Model diagnostics showed no major violations of linear regression assumptions. Residuals were approximately normally distributed, no influential observations were identified, and variance inflation factors indicated no problematic multicollinearity.

(Results summarized in Table 3-7 and illustrated in Figure 4.)

3.5 Predictive Performance

When evaluated on a held-out test set, the final model showed moderate predictive performance, with a RMSE of 15.3 and a MAE of 11.4.

4 Discussion

This analysis examined the relationship between prenatal and perinatal substance exposures and childhood cognitive performance. The results show that maternal smoking, prenatal opioid exposure, and prenatal cannabis exposure are independently associated with lower cognitive scores, directly addressing the research question.

Maternal smoking both before and during pregnancy remained significant in multivariable and stepwise-selected models, indicating robust associations not fully explained by co-occurring exposures. Prenatal opioid and cannabis exposures also showed independent negative associations. In contrast, prenatal alcohol exposure and biospecimen-based markers were not independently associated with cognitive score after adjustment, underscoring the importance of multivariable modeling when exposures are correlated.

Statistically, the modest variance explained by the final models is consistent with the multifactorial nature of cognitive development. Agreement between AIC- and BIC-based model selection supports the stability of the identified predictors, while collinearity among exposures—particularly involving the poly-substance index—necessitated careful model specification to ensure identifiability.

Key limitations include the use of complete-case analysis, binary exposure measures without dose or timing information, and the cross-sectional design, which precludes causal inference. Residual confounding cannot be ruled out.

Future work could incorporate more detailed exposure characterization, longitudinal designs, and methods better suited for correlated exposures to further clarify these associations.

While the selected model provides interpretable associations, its predictive performance was modest, highlighting the distinction between explanatory and predictive modeling goals. A Bayesian re-estimation did not improve RMSE/MAE, suggesting that predictive accuracy is primarily limited by the available predictors rather than the estimation framework.

References

England, L. J., Bennett, C., Denny, C. H., Honein, M. A., Gilboa, S. M., Kim, S. Y., Guy, G. P., Tran, E. L., Rose, C. E., Bohm, M. K., & Boyle, C. A. (2020). Alcohol use and co-use of other substances among pregnant females aged 12–44 years — United States, 2015–2018. *MMWR. Morbidity and Mortality Weekly Report*, 69(31), 1009–1014.

Argote, M., Hilson, L., Sorkhou, M., & Rabin, R. A. (2025). A systematic review investigating prenatal cannabis and tobacco co-exposure: Impacts on neonatal, behavioral, cognitive and physiological outcomes. *Drug and Alcohol Dependence Reports*, 17, 100376.

Appendix A

A.1 Key Programming Code with Comments

Code 1 Data Preparation and Variable Construction

```
# Load core packages
library(tidyverse)

# Read data
hbcd <- read.csv("hbcd_simulated.csv")

# Select outcome and exposure variables
dat0 <- hbcd %>%
  select(cog_score, smoking_prepreg, smoking_preg, alcohol_preg,
    prenatal_opioid, prenatal_cannabis, urine_opioid_pos, urine_cannabis_pos,
    cord_cotinine, poly_substance_index)

# Convert binary exposures to numeric (0/1)
binary_vars <- c("smoking_prepreg", "smoking_preg", "alcohol_preg",
  "prenatal_opioid", "prenatal_cannabis", "urine_opioid_pos", "urine_cannabis_pos")

dat0 <- dat0 %>%
  mutate(across(all_of(binary_vars), as.numeric))

# Log-transform cord blood cotinine
dat0 <- dat0 %>%
  mutate(cord_cotinine_log = log(cord_cotinine + 0.1))

# Complete-case dataset
dat <- dat0 %>%
```

```
  drop_na(cog_score, smoking_prepreg, smoking_preg,
    alcohol_preg, prenatal_opioid, prenatal_cannabis, urine_opioid_pos,
    urine_cannabis_pos, cord_cotinine_log, poly_substance_index)
```

Code 2 Univariate Linear Regression Analyses

```
library(broom)

# Function for univariate regression
run_univariate <- function(data, exposure, is_binary = TRUE) {
  if (is_binary) {
    formula <- as.formula(paste0("cog_score ~ factor(", exposure, ")"))
  } else {
    formula <- as.formula(paste0("cog_score ~ ", exposure))
  }

  lm(formula, data = data) %>%
    tidy(conf.int = TRUE)
}

# Binary exposures
binary_exposures <- binary_vars

# Continuous exposures
continuous_exposures <- c("cord_cotinine_log", "poly_substance_index")

# Run univariate models
uni_binary <- map_dfr(binary_exposures,
                      ~ run_univariate(dat, .x, TRUE))
uni_continuous <- map_dfr(continuous_exposures,
                          ~ run_univariate(dat, .x, FALSE))
```

Code 3 Multivariable Regression and Model Selection

```
library(MASS)
library(car)

# Full multivariable model (component-based)
model_full <- lm(
  cog_score ~
    factor(smoking_prepreg) + factor(smoking_preg) + factor(alcohol_preg) +
    factor(prenatal_opioid) + factor(prenatal_cannabis) + factor(urine_opioid_pos) +
    factor(urine_cannabis_pos) + cord_cotinine_log, data = dat)
```

```
# Stepwise model selection
model_aic <- stepAIC(model_full, direction = "backward", k = 2)
model_bic <- stepAIC(model_full, direction = "backward", k = log(nrow(dat)))

anova(model_aic, model_full)
anova(model_bic, model_full)

# Compare retained predictors
terms_aic <- names(coef(model_aic))[-1]
terms_bic <- names(coef(model_bic))[-1]
intersect(terms_aic, terms_bic)
```

Code 4 Model Diagnostics

```
# Residual diagnostics
par(mfrow = c(2, 2))
plot(model_bic)

# Multicollinearity
vif(model_bic)
cor(dat[,-1])

# Heteroscedasticity test
library(lmtest)
bptest(model_bic)
```

Code 5 Predictive Modeling – MLR

```
# Predictive Modeling (Train/Test Split)

set.seed(123)

# Split data into training (70%) and test (30%) sets
n <- nrow(dat)
train_index <- sample(seq_len(n), size = 0.7 * n)

train_data <- dat[train_index, ]
test_data  <- dat[-train_index, ]

# Fit predictive model on training data
# Model structure matches the final explanatory model
model_pred <- lm(
  cog_score ~
    factor(smoking_prepreg) +
```

```
    factor(smoking_preg) +
    factor(prenatal_opioid) +
    factor(prenatal_cannabis),
  data = train_data
)

# Predict cognitive score in test data
pred_test <- predict(model_pred, newdata = test_data)

# Evaluate predictive performance for MLR model
rmse <- sqrt(mean((test_data$cog_score - pred_test)^2))
mae  <- mean(abs(test_data$cog_score - pred_test))
```

Code 6 Predictive Modeling – Bayesian

```
# Bayesian linear regression with conjugate prior:
#   beta ~ N(0, tau2 * I)
#   sigma2 ~ Inv-Gamma(a, b)

library(MASS)   # for mvrnorm

# Reuse the same train/test split created in Code 5:
# train_data, test_data already exist

# Use the same predictors as the final predictive model.
# Since these exposures are coded 0/1 in dat, treating them as numeric
# yields the same fitted values as factor( ) indicator coding.
form_bayes <- cog_score ~ smoking_prepreg + smoking_preg + prenatal_opioid +
prenatal_cannabis

# Design matrices
X_train <- model.matrix(form_bayes, data = train_data)
y_train <- train_data$cog_score

X_test  <- model.matrix(form_bayes, data = test_data)
y_test  <- test_data$cog_score

# Priors (weakly informative; similar to lecture example)
tau2 <- 10000
a <- 1
b <- 1

# Helper: Inv-Gamma sampler using Gamma relationship
# If U ~ Gamma(shape, rate), then 1/U ~ Inv-Gamma(shape, rate)
rinvgamma_base <- function(n, shape, rate) {
  1 / rgamma(n, shape = shape, rate = rate)
}
```

```
# Gibbs sampler settings
set.seed(123)
n_iters <- 10000
burn <- 5000

n <- nrow(X_train)
p <- ncol(X_train)

Beta <- matrix(0, nrow = n_iters, ncol = p)
sigma2 <- numeric(n_iters)
sigma2[1] <- 1

for (iter in 2:n_iters) {

  # ---- Sample beta | sigma2, y, X ----
  V_beta <- solve( (t(X_train) %*% X_train) / sigma2[iter - 1] + diag(p) / tau2 )
  m_beta <- V_beta %*% (t(X_train) %*% y_train) / sigma2[iter - 1]
  Beta[iter, ] <- mvrnorm(n = 1, mu = as.vector(m_beta), Sigma = V_beta)

  # ---- Sample sigma2 | beta, y, X ----
  resid <- y_train - X_train %*% Beta[iter, ]
  shape_post <- n / 2 + a
  rate_post  <- as.numeric(t(resid) %*% resid) / 2 + b
  sigma2[iter] <- rinvgamma_base(1, shape = shape_post, rate = rate_post)
}

post_idx <- (burn + 1):n_iters

# Posterior summaries for coefficients
beta_post_mean <- colMeans(Beta[post_idx, , drop = FALSE])
beta_post_ci <- apply(Beta[post_idx, , drop = FALSE], 2, quantile, probs = c(0.025, 0.975))

beta_post_mean
beta_post_ci

# Point prediction: posterior mean of E[y|x,beta]
pred_bayes <- as.vector(X_test %*% beta_post_mean)

# Evaluate predictive performance (compare to OLS Code 5)
rmse_bayes <- sqrt(mean((y_test - pred_bayes)^2))
mae_bayes  <- mean(abs(y_test - pred_bayes))

rmse_bayes
mae_bayes

# Optional: Posterior predictive intervals (uncertainty for new outcomes)
# Use a subset of posterior draws to keep memory reasonable
set.seed(123)
draws <- sample(post_idx, size = 1000)
```

```
mu_draws <- X_test %*% t(Beta[draws, , drop = FALSE])
eps_draws <- sweep(
  matrix(rnorm(nrow(X_test) * length(draws)), nrow(X_test), length(draws)),
  2, sqrt(sigma2[draws]), "*"
)
yrep_draws <- mu_draws + eps_draws

pred_PI <- t(apply(yrep_draws, 1, quantile, probs = c(0.025, 0.975)))
head(pred_PI)

# Evaluate predictive performance for Bayesian model
rmse_bayes <- sqrt(mean((y_test - pred_bayes)^2))
mae_bayes  <- mean(abs(y_test - pred_bayes))
```

A.2 Brief Analysis Log

During the analysis, several alternative modeling strategies were explored before selecting the final models reported in the main text. Initial multivariable models included both individual substance exposure indicators and a poly-substance exposure index. This specification resulted in non-identifiable coefficients due to perfect collinearity, reflecting the deterministic relationship between the index and its component variables. As a result, component-based and index-based models were evaluated separately rather than simultaneously.

Models incorporating biospecimen-based exposure measures were also examined. Although these variables showed associations with cognitive score in univariate analyses, their effects were attenuated and no longer statistically significant after adjustment for correlated questionnaire-based exposures. This comparison highlighted the importance of multivariable adjustment when multiple measures capture overlapping exposure information.

Both AIC- and BIC-based stepwise selection procedures were applied to the full component-based model. The convergence of these criteria on the same set of predictors increased confidence in the stability of the selected model. The final model was chosen based on consistency across selection criteria, statistical identifiability, and interpretability, rather than on goodness-of-fit alone.

In addition, a predictive version of the final model was evaluated using a train–test split to assess out-of-sample performance, highlighting the distinction between explanatory and predictive modeling goals.

A **Bayesian** predicted model was also fitted. However, posterior mean coefficient estimates were very similar to the OLS estimates, and the out-of-sample predictive accuracy was essentially unchanged (in Table 6, test RMSE ≈ 15.3, MAE ≈ 11.4, matching the frequentist baseline). Given no improvement in RMSE/MAE and the goal of maintaining straightforward interpretability for the main report, the predictive model was retained as the primary prediction approach, while the Bayesian implementation is included in the Appendix for completeness and uncertainty quantification.

A.3 Figures and Tables

Fig. 1 Distribution of missing data

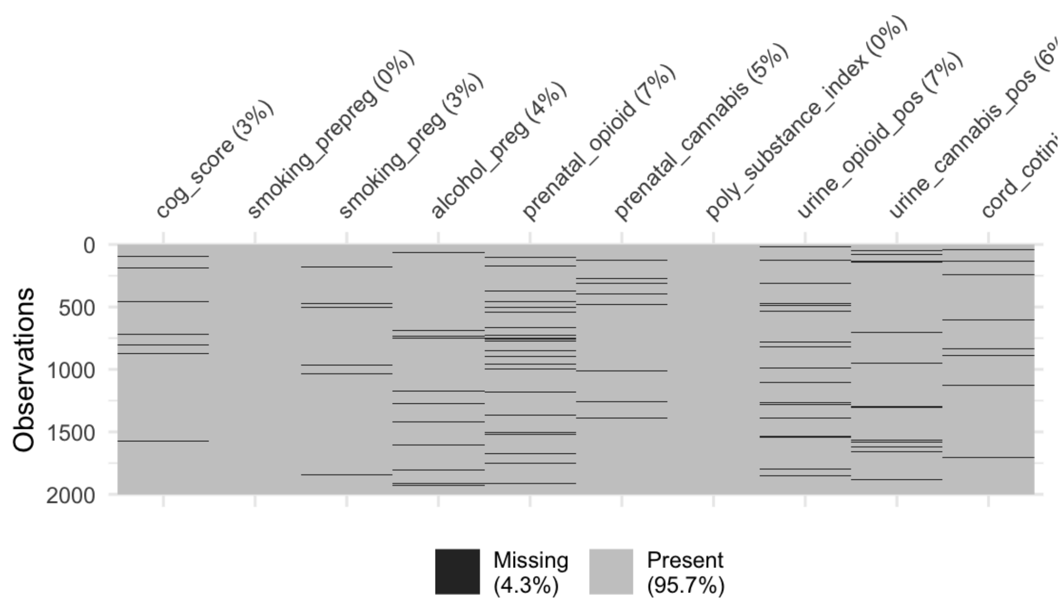


Fig. 2 Distribution of *cog_score* and *cord_cotinine*

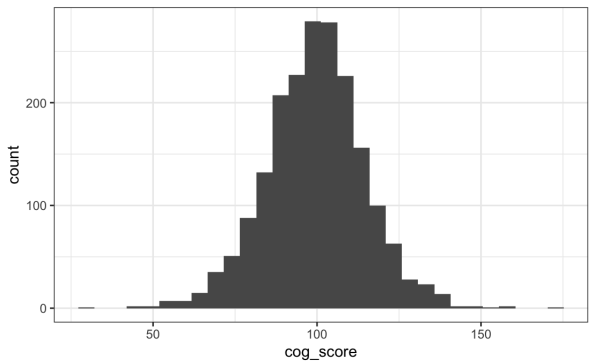
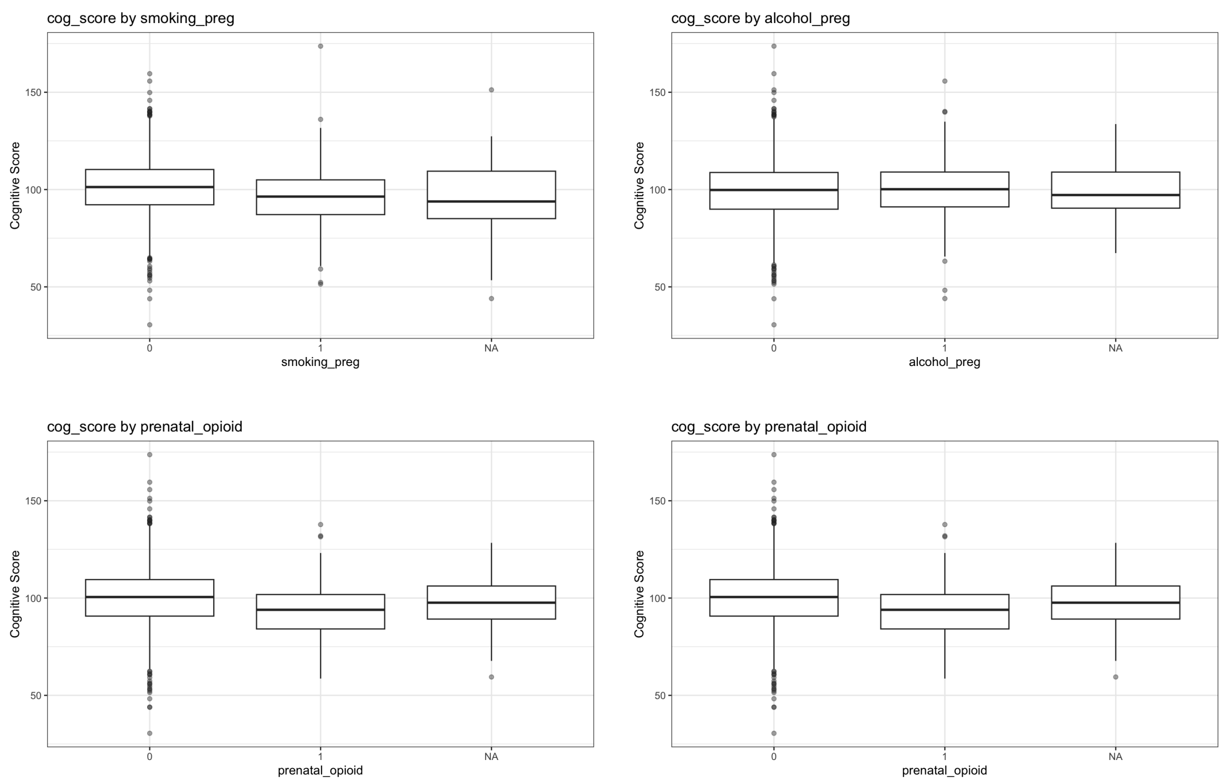
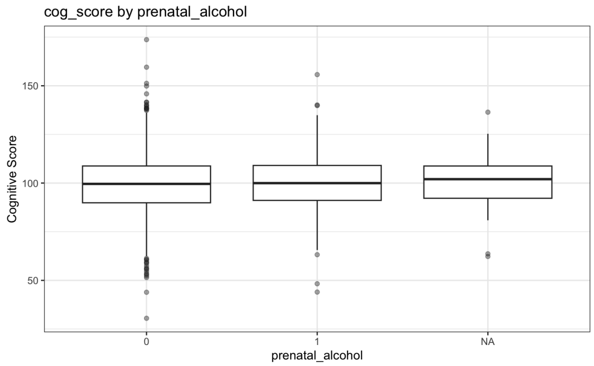


Fig. 3 Distribution of cognitive scores by prenatal and perinatal substance exposures





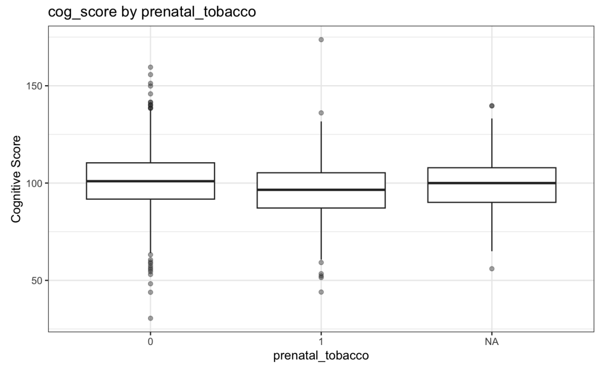


Table 1 Univariate analysis

exposure	term_type	estimate	p.value	significant	CI
alcohol_preg	Intercept	99.994	0.00E+00	Yes	[99.123, 100.866]
	Exposure	1.196	3.00E-01	No	[-1.066, 3.458]
cord_cotinine_log	Intercept	100.4	0.00E+00	Yes	[99.589, 101.21]
	Exposure	-0.675	4.30E-04	Yes	[-1.05, -0.3]
poly_substance_index	Intercept	102.419	0.00E+00	Yes	[101.347, 103.491]
	Exposure	-3.509	1.28E-09	Yes	[-4.635, -2.383]
prenatal_cannabis	Intercept	100.818	0.00E+00	Yes	[99.948, 101.689]
	Exposure	-4.159	2.30E-04	Yes	[-6.368, -1.951]
prenatal_opioid	Intercept	100.571	0.00E+00	Yes	[99.737, 101.406]
	Exposure	-5.113	8.01E-04	Yes	[-8.098, -2.128]
smoking_preg	Intercept	101.491	0.00E+00	Yes	[100.568, 102.415]
	Exposure	-5.109	4.18E-08	Yes	[-6.925, -3.292]
smoking_prepreg	Intercept	101.316	0.00E+00	Yes	[100.359, 102.274]
	Exposure	-3.765	2.26E-05	Yes	[-5.503, -2.028]
urine_cannabis_pos	Intercept	100.632	0.00E+00	Yes	[99.743, 101.521]
	Exposure	-2.501	1.80E-02	Yes	[-4.573, -0.43]
urine_opioid_pos	Intercept	100.508	0.00E+00	Yes	[99.655, 101.361]
	Exposure	-2.954	2.21E-02	Yes	[-5.483, -0.425]

Table 2 Multivariate analysis for the full model

term	estimate	std.error	p.value	significant	CI
(Intercept)	103.291	0.6226483	0.00E+00	Yes	[102.069, 104.512]
factor(smoking_prepreg)1	-3.036	0.8958868	7.22E-04	Yes	[-4.794, -1.279]
factor(smoking_preg)1	-5.311	1.2121977	1.28E-05	Yes	[-7.689, -2.933]
factor(alcohol_preg)1	0.642	1.1292765	5.69E-01	No	[-1.573, 2.858]
factor(prenatal_opioid)1	-5.793	2.0545085	4.88E-03	Yes	[-9.823, -1.762]
factor(prenatal_cannabis)1	-4.308	1.5371592	5.15E-03	Yes	[-7.323, -1.292]
factor(urine_opioid_pos)1	0.425	1.7381208	8.07E-01	No	[-2.985, 3.835]
factor(urine_cannabis_pos)1	0.437	1.43744	7.61E-01	No	[-2.383, 3.257]
cord_cotinine_log	0.218	0.2517689	3.88E-01	No	[-0.276, 0.712]

Table 3 Multivariate analysis for the final model

term	estimate	std.error	p.value	significant	CI
(Intercept)	103.302	0.5559362	0.00E+00	Yes	[102.211, 104.393]
factor(smoking_prepreg)1	-2.921	0.8811127	9.40E-04	Yes	[-4.65, -1.193]
factor(smoking_preg)1	-4.655	0.9255842	5.63E-07	Yes	[-6.47, -2.839]
factor(prenatal_opioid)1	-5.392	1.4916303	3.12E-04	Yes	[-8.318, -2.466]
factor(prenatal_cannabis)1	-3.98	1.1052765	3.29E-04	Yes	[-6.148, -1.812]

Table 4 Multivariate analysis for the Bayesian model

term	posterior mean	std.error	CrI
(Intercept)	103.245	0.6550599	[101.955, 104.513]
factor(smoking_prepreg)1	-2.700	1.0097444	[-4.6693, -0.708]
factor(smoking_preg)1	-4.071	1.1102261	[-6.262, -1.890]
factor(prenatal_opioid)1	-6.470	1.7415356	[-9.818, -3.098]
factor(prenatal_cannabis)1	-3.588	1.2549622	[-5.997, -1.141]

Table 5 Model Selection

Model	**df**	**AIC**	**BIC**	**Factor**	**VIF**
model_aic	6	10570.88	10601.87	*smoking_prepreg*	1.026505
				smoking_preg	1.025793
				prenatal_opioid	1.001343
				prenatal_cannabis	1.002359
model_bic	6	10570.88	10601.87	*smoking_prepreg*	1.026505
				smoking_preg	1.025793
				prenatal_opioid	1.001343
				prenatal_cannabis	1.002359
model_full	10	10577.64	10629.29	*smoking_prepreg*	1.058935
				smoking_preg	1.755658
				alcohol_preg	1.005581
				prenatal_opioid	1.895577
				prenatal_cannabis	1.934570
				urine_opioid_pos	1.898409
				urine_cannabis_pos	1.935249
				cord_cotinine_log	1.801972

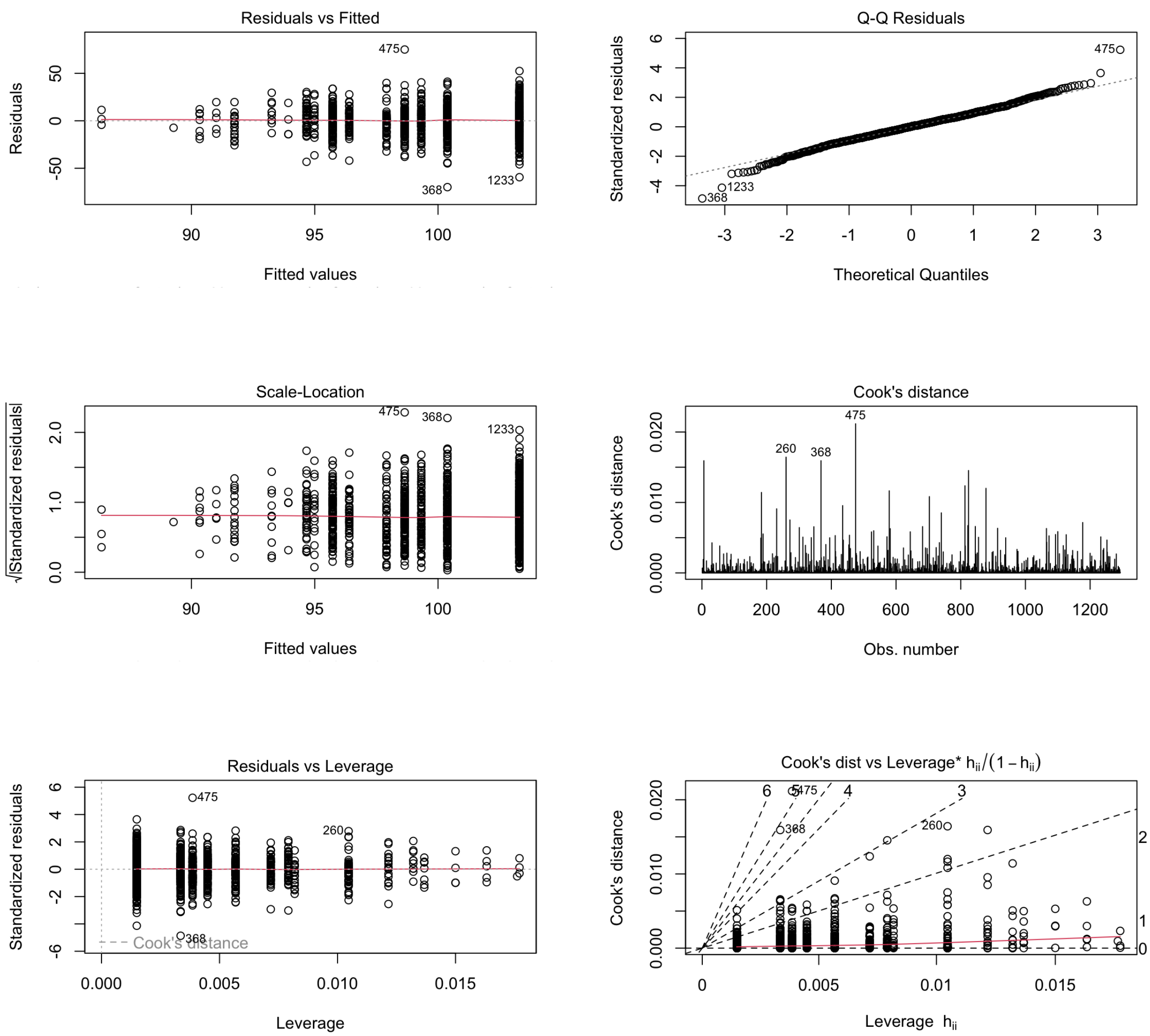
Table 6 Model Selection

	RMSE	MAE
model_aic	15.33458	11.38664
model_bic	15.33458	11.38664
fit_bayes	15.3344	11.38668

Table 7 Pearson correlations

Variable	*smoking_ prepreg*	*smoking_preg*	*alcohol_preg*	*prenatal_ opioid*	*prenatal_ cannabis*	*urine_opioid_ pos*	*urine_ cannabis_pos*	*cord_cotinine*	*poly_substance _index*	*cord_ cotinine_log*
smoking_ prepreg	1.0000	0.1555	-0.0537	0.0082	0.0413	0.0441	0.0376	0.2299	0.0939	0.2223
smoking_preg	0.1555	1.0000	-0.0427	-0.0269	0.0199	-0.0165	0.0161	0.8392	0.5997	0.6548
alcohol_preg	-0.0537	-0.0427	1.0000	0.0081	-0.0351	0.0080	-0.0131	-0.0415	0.4632	-0.0399
prenatal_ opioid	0.0082	-0.0269	0.0081	1.0000	-0.0215	0.6856	-0.0118	-0.0120	0.3572	0.0114
prenatal_ cannabis	0.0413	0.0199	-0.0351	-0.0215	1.0000	-0.0057	0.6940	0.0390	0.5008	0.0332
urine_opioid_ pos	0.0441	-0.0165	0.0080	0.6856	-0.0057	1.0000	0.0248	0.0011	0.2520	0.0028
urine_ cannabis_pos	0.0376	0.0161	-0.0131	-0.0118	0.6940	0.0248	1.0000	0.0334	0.3558	0.0331
cord_cotinine	0.2299	0.8392	-0.0415	-0.0120	0.0390	0.0011	0.0334	1.0000	0.5159	0.8405
poly_substance _index	0.0939	0.5997	0.4632	0.3572	0.5008	0.2520	0.3558	0.5159	1.0000	0.4081
cord_ cotinine_log	0.2223	0.6548	-0.0399	0.0114	0.0332	0.0028	0.0331	0.8405	0.4081	1.0000

Fig. 4 Model Diagnostics



Appendix B

B.1 Tools used

This project benefited from the use of ChatGPT for debugging and organizing R code, and improving the clarity and conciseness of written text.

B.2 Example prompts

Example Prompt 1

“Can you help me check whether my R code for univariate and multivariable linear regression is structured correctly and identify any potential syntax or workflow issues?”

AI Response and Evaluation:

The AI reviewed the structure of the R code and suggested improvements related to code organization and variable handling. These suggestions helped streamline the workflow but did not alter the statistical models themselves. All recommendations were manually checked in R, and only those consistent with the course material were adopted.

Example Prompt 2:

“Help me compress the Results and Discussion sections to meet strict word limits while preserving statistical meaning.”

AI Response and Evaluation:

The AI provided shortened draft versions of the text that preserved key findings and interpretations. These drafts were reviewed and manually edited to ensure accuracy, appropriate emphasis, and consistency with the actual statistical results. Any wording that risked overstating conclusions or implying causality was corrected by the author.